SHIFANA R

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GitHub | LinkedIn | LeetCode

Education

K. Ramakrishnan College of Engineering 2023 – 2027
B.E. in Computer Science and Engineering – CGPA: 8.25/10

Brindhavan Higher Secondary School, Pattukkottai, TN 2023
HSC – 87.33%

St. Isabel's Girls' Higher Secondary School, Pattukkottai, TN 2021
SSLC – 100%

Experience

Product Engineering Intern Mar. 2026 – Jun. 2026
Matrimony.com Limited *Chennai, India*

- Resolved 25+ production defects across Android (Kotlin) and web (Angular) applications, improving onboarding and profile management workflows.
- Optimized client-side state handling and GraphQL integrations in Angular, reducing unnecessary re-renders and improving UI responsiveness.
- Streamlined Android build and deployment workflows across 118+ app variants using Gradle and CI/CD pipelines, reducing manual effort across regional Community Matrimony apps.

GenAI Intern Sep. 2025 – Nov. 2025
Kactii *Remote*

- Built containerized RAG services and multi-step reasoning agents using **Python**, **FastAPI**, **LangChain**, **LangGraph**, and **Docker**, implementing tool-calling and optimized vector retrieval.
- Contributed to **KactLabs** open-source projects, developing modular agent tool connectors and core features for **Toksum** alongside developer documentation.

Projects

High-Performance Persistent Key-Value Store | *C++17, STL, Multithreading, File I/O, CMake*

- Developed a thread-safe key-value storage engine in C++17 supporting PUT, GET, and DELETE operations, with file-based persistence to retain data across application restarts.
- Designed a Write-Ahead Log (WAL) for recording data mutations before applying them, enabling recovery of stored operations after an unexpected shutdown.
- Implemented LRU caching, TTL-based key expiration, and background log compaction to manage frequently accessed data, expire stale entries, and reduce storage overhead.

NexusQuery AI | *Python, FastAPI, React, FAISS, BM25, MySQL*

- Built an enterprise knowledge assistant using Retrieval-Augmented Generation (RAG) to query documents, GitHub repositories, internal wikis, and web pages through natural-language questions.
- Implemented hybrid retrieval using FAISS vector search, BM25 keyword ranking, and Reciprocal Rank Fusion (RRF), with Sentence Transformers embeddings and source-level citations.
- Developed FastAPI REST endpoints and a React frontend, integrating document ingestion, streaming responses via Server-Sent Events (SSE), JWT authentication, and role-based access control.

Technical Skills

Core CS: DSA, OOP, OS, DBMS, Networks, Multithreading
Languages: C++, C, Python, SQL, JavaScript
Systems: Linux, System Calls, Synchronization, Memory Management, STL, Smart Pointers, RAII
Networking: TCP/IP, UDP, DNS, HTTP/HTTPS, Sockets
Tools: Git, GitHub, CMake, CI/CD
Development/AI: Node.js, FastAPI, REST APIs, Angular, React, LangChain, LangGraph, RAG, FAISS

Certifications

AWS Certified Solutions Architect – Associate (SAA-C03)
Introduction to Model Context Protocol – Anthropic

Achievements & Patents

Selected as 1 of 15 interns from 8,000+ applicants for the Matrimony.com Product Engineering Internship.
Awards: NTSE District Rank 1 (2022) among students from government and aided schools; NMMS Scholarship recipient (2019).
Hackathons: National Finalist, TechNova AI; Finalist, Yukti.
Leadership: Student Member, IIC and ICC.
Patents: Fingerprint-Based PIN Authentication (20 2025 100 969.2); AI-Synthesized Quantum-Dot Photoelectrochemical Mat (202541084924); Bio-Inspired Foam Dust for Space Debris Management (202541084884).